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Water-mediated hydrogen bonds and local side-chain interactions in the cooperative collapse and expansion of PNIPAM oligomers

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Poly(N-isopropylacrylamide) (PNIPAM), a thermoresponsive homopolymer, is a well-established model for investigating coil-to-globule transitions. Here, we combine long molecular dynamics (MD) simulations, data sonification, and graph-theory analysis to elucidate the roles of intramolecular and PNIPAM–solvent hydrogen-bond (H-bond) patterns in the PNIPAM globule–coil equilibrium. Our analysis separates the driving forces for compaction into two contributions: the entropic gain from the loss of hydration water around hydrophobic patches and the enthalpic stabilization from water H-bonded to PNIPAM. We find that the role of the solvent in polymer compaction is more active and complex than has been previously assumed. Our observations indicate that direct, intrachain hydrogen bonds between amide groups (N–H···O=C) are not the primary stabilizing force. Instead, the collapsed globule contains an N–H···N network of local side-chain interactions and is stabilized by a dynamic network of persistent, long-distance water bridges, where individual water molecules form hydrogen bonds with multiple parts of the polymer chain.

sonification | bound water | wrap water | polymer compactification | water-mediated interactions

Significance

Materials that change shape with temperature are foundational for applications ranging from controlled drug release to soft robotics. Through computational simulations of how a model polymer collapses in water, we show that water is an active participant in this process—forming structural water bridges that pull the polymer into its compact state. The method developed here for analyzing hydrogen bond dynamics using auditory analytics—an approach that integrates classic statistical analyses with data sonification and visualization—provides critical insights into the complex dynamics typical of high-dimensional chemical and biological systems.

Poly(N-isopropylacrylamide), or PNIPAM, is a homopolymer frequently employed in the synthesis of temperature-sensitive hydrogels with tunable mechanical properties (1). It is often used as a model system for studying coil-to-globule transitions at a lower critical solution temperature (LCST) (2). A simple picture for polymer compaction upon heating is the increasing entropic driving force at higher temperature (3), leading to the release of hydration water around hydrophobic patches (4). Thus, the transition is sensitive to solvent composition (5, 6) However, hydrogen bonding also plays an important role in the formation of compact polymer globules and their structural organization (7).

Aqueous PNIPAM has been extensively investigated by experiment and molecular dynamics (MD) simulation. The latter work has led to improvements in simulated transition temperatures (8), confirmed that PNIPAM tacticity has only a small effect on globule structure (9), measured diffusion of water within the polymer matrix (10), and looked at the effects of osmolytes (11) among many other interesting properties. In a classic study, Tamai, Tanaka, and Nakanishi studied hydrogen bonding of water near the PNIPAM surface (12); since that time however, the mechanism by which these interactions drive the transition from extended to compact states above the LCST has remained largely unexplored.

Here, we use long MD simulations (cumulative 16 μs) to study the changes in hydrogen bonding during the cooperative passages of PNIPAM through the transition state ensemble as the oligomer switches between extended coil and compact globule states. We investigate a 30-mer and a 60-mer of NIPAM in aqueous solution in equilibrium near the transition temperature, as it has been shown that cooperativity occurs for sizes <60 kDa (<539 subunits) (13). We are able to observe multiple rapid coil–globule and globule–coil transitions in the simulation, as well as a population of long-lived intermediate compactness states. Since PNIPAM lacks sequence-specific cooperativity, not all of its transitions are fast relative to its dwell times. However, we focus here on the subset of cooperative transitions that occur between structurally distinct states.

We combined interactive data sonification and visualization tools with a 2D reaction coordinate state space derived via dimensionality reduction to track complex hydrogen-bonding dynamics. Potentially interesting patterns were followed up by rigorous modeling and statistical analysis. With roots in early tools like the Geiger counter in 1908 (14), modern use of scientific data sonification has paralleled the expansion of scientific

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Competing interest statement: C.S. and K.J.H. are co-founders of Symbolic Sound Corporation, developers of the Kyma language used in this work. The other authors declare no competing interests.

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computing and visualization (15–17). Past protein and nucleic acid studies have focused on mapping residue sequences and structure-property correlations to temporal auditory events (18–20); ref. 19 uses normal modes to create a physics-based sonification for each amino acid assigned to a piano roll. Our recent work focuses on the dynamics of hydrogen bond formation during protein folding transitions (21). Here, we extend this approach to analyze hydrogen bond dynamics during the coil-to-globule transition of the synthetic polymer PNIPAM.

The influence of direct polymer–polymer hydrogen bonds (DBs) on the dynamics of the polymer chain is less significant (22) than had been widely assumed. We find instead that the efficient collapse of PNIPAM is enabled by a combination of water-mediated hydrogen bonds (WBs) and amide–amide side-chain interactions (SCBs characterized by a parallel N–H/N–H structure), with most transition state passages in the 2 to 3 ns range. We find, as expected, the duration of transition state passages grows with polymer size. DB and WB H-bonding frequently traps PNIPAM in states of intermediate compactness that persist for up to 100 s of ns. As expected from the force field-dependent persistence length of PNIPAM (1 to 3 nm) (23), we observe compact states with one to five kinks in the polymer chain. We investigated correlations between DBs, WBs, and SCBs before, during, and after transition state passage between globule and coil forms of the PNIPAM oligomer.

Results

Transitions Between Coil and Globule States. Our goal is to elucidate the hydrogen bonding and structural arrangements that occur when PNIPAM makes rapid cooperative transitions between collapsed globule and extended coil forms. To that end, we investigated the conformational transitions of a single PNIPAM 30-mer in water by performing 11 independent MD simulations totaling about 16 μs (Fig. 1 and *SI Appendix*, Fig. S1). The simulations employed a modified OPLS-AA force field (8, 24) for PNIPAM in combination with SPC/E water (25). Systems were simulated in the NPT ensemble at 1 atm and 310 K using the Nosé–Hoover thermostat, with a cubic box of approximately $10 \times 10 \times 10$ nm^3 to ensure sufficient bulk solvent and to avoid boundary effects where the polymer could interact with itself. To investigate the influence of chain length, we also simulated a 60-mer PNIPAM in a $12.5 \times 12.5 \times 12.5$ nm^3 box, maintaining the same force field, water model, and temperature/pressure conditions (*SI Appendix*, Fig. S2). See *Methods* and *SI Appendix*, Methods for more details.

We modeled the multimodal distribution of the radius of gyration (R_g) as a Gaussian Mixture Model (GMM) comprising two to four weighted Gaussian distributions (Fig. 1*C* and *SI Appendix*, Table S1 and Fig. S3). This approach is motivated by the principle that random walk ensembles of sufficiently long chains (here ≥30 monomers)

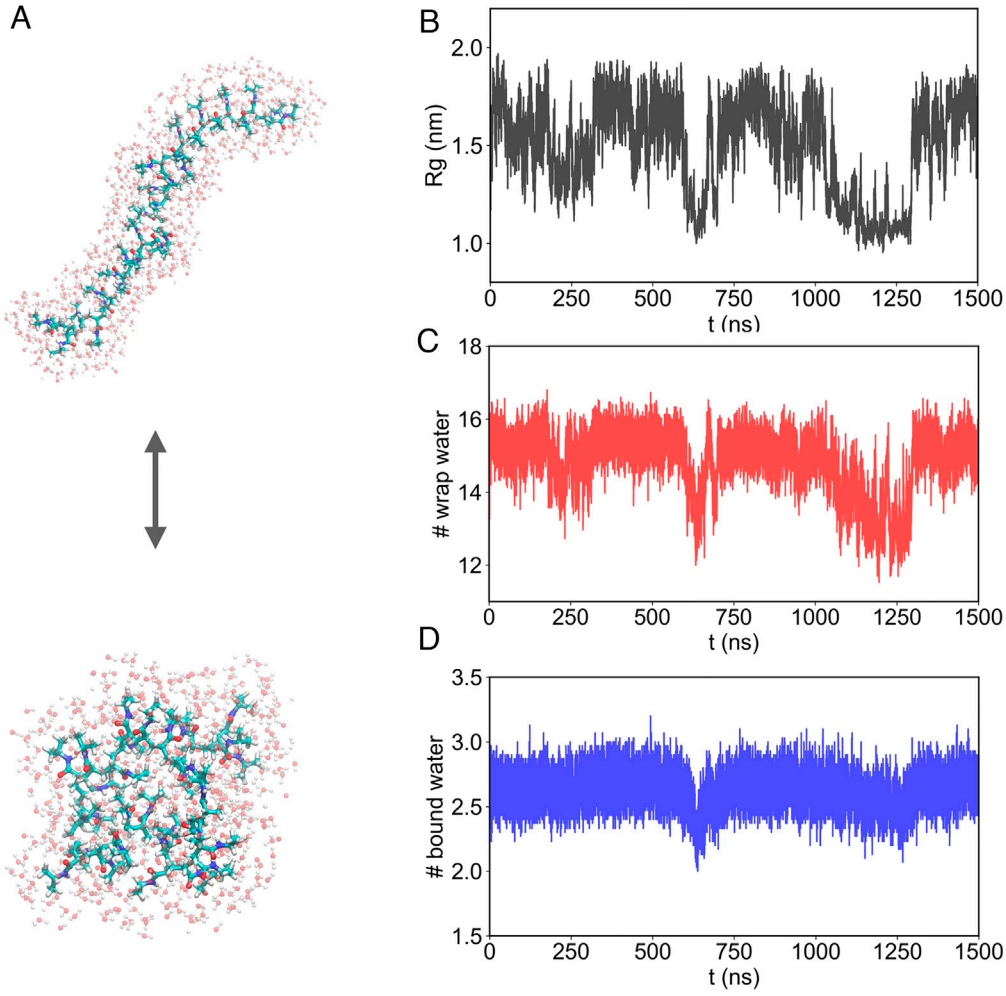


Fig. 1. (*A*) Example of an extended (*Top*) and compact (*Bottom*) structure of PNIPAM. (*B*) Radius of gyration (R_g)trajectory for one of the 11 simulation trajectories. (*C*) Radius of gyration distribution (blue) and fit (red) to 3 Gaussian distributions, shown in green. (*D*) Number of "Wrap" water molecules for the trajectory in (*B*) E. Number of "Bound" water molecules for the trajectory in *B*. *SI Appendix*, Fig. S1 shows a concatenation of all 11 trajectories and transition regions.

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have approximately Gaussian distributed sizes (26). Not unexpectedly, for the PNIPAM 30-mer, three states—we call them globule, intermediate, and coil as size increases—provide an adequate fit of the data, except for the largest $R_g > 1.85$ nm, which are sampled very rarely by the simulation. Both 30-mer and 60-mer have a similar globule population (6 to 7%), but the coil becomes more populated in the 60-mer (64% up from 37%) at the expense of the intermediate (30% down from 56%) so the larger oligomer is more cooperative, likely due to chain stiffness (27) the intermediate ensemble populations suggest that PNIPAM's homopolymer collapse cannot be adequately described by a simple two-state model. In our three-state model, the globule and coil probabilities intersect at about 1.3 nm for the PNIPAM 30-mer, in agreement with a previous simulation study of PNIPAM 40-mer (8) and a study of PNIPAM 30-mer where the globular state was defined as $R_g < 1.2$ nm (28).

To histogram passage time durations of coil–globule transitions, we developed a Kinetic Transition Estimator algorithm (KiTE) and compared its results for consistency with the more computationally expensive Step Transition and State Identification (STaSI) algorithm and logistic equation fits (29, 30). Direct passages between 30-mer globule and coil assigned by the KiTE and STaSI estimators are shown in *SI Appendix*, Fig. S1. For both methods, the histogram of passage times peaks at 2.5 ± 0.5 ns with a tail to ≈10 ns, defining the "speed limit" for full PNIPAM collapse (see histograms in *SI Appendix*, Fig. S4 and Tables S2 and S3). For the 60-mer, we find (with more limited sampling) a similar peak at 2.5 ± 0.5 ns and a more pronounced tail to ≈20 ns (*SI Appendix*, Fig. S5). The slowing of some passages with polymer size, although defined by only two points, is consistent with a size scaling between the scaling exponents of 4/3 and 2 expected for a hydrodynamic vs. a simple Brownian heat bath (31). These passage times are similar to experimentally measured contact times for glycine oligomers of similar length (~5 ns) (32), so the two systems likely have a similar number of rotameric degrees of freedom per residue (two for the backbone, one or two for the side chain).

PNIPAM shows some signs of cooperativity. For example, the 30-mer has a mean dwell time (77 ns) in states that is roughly 30 times longer than the most likely passage time (2.5 ns). Similar results (with more limited sampling) are obtained for the 60-mer. Nonetheless, as a homopolymer with a glassy energy landscape, in addition to featuring direct coil–globule transitions, PNIPAM also collapses into or escapes from many partially compact intermediates whose lifetimes exceed 20 ns, the tail of the passage time distributions. The oligomer must escape from such "traps" before undergoing full collapse or expansion (Fig. 1*B*). STaSI (*Methods*) provided additional confirmation that the trajectories are adequately described by a three-state model (the intermediate is the middle state in *SI Appendix*, Figs. S1*B* and S2). These semicompact trap states are consistent with the multimodal R_g distribution (Fig. 1*C*).

Conventional autocorrelation analysis also revealed multiple, distinct time scales in the trajectory data. *SI Appendix*, Fig. S6 shows the autocorrelations for the 30-mer and the 60-mer on a log–log plot, revealing separation into a "fast" and a "slow" time scale; these scales correspond, respectively, to transition state passage and dwell times in compact, intermediate, or extended states. The fact that these time scales are consistent with those obtained from the KiTE and STaSI analyses supports a description of PNIPAM collapse and expansion as transitions over small free energy barriers.

PNIPAM Solvation. To study the water dynamics and hydrogen bonding, we categorized the water molecules in PNIPAM's first solvation shell as either "Wrap" water or "Bound" water (*Methods* and *SI Appendix*, Table S4). A classification previously established through Terahertz spectroscopy and MD simulations of solvated polymers (4), Wrap water forms a hydrogen-bonded network around nonpolar regions without directly H-bonding to the polymer (3), thus forming a two-dimensional water–water H-bond network wrapped around the solute, while Bound water hydrogen-bonds to the polymer's polar groups. Wrap and Bound waters differ in their residence time (mean time between entering and leaving the PNIPAM first solvation shell) (*SI Appendix*, Fig. S7): Wrap waters, which lack the extensive H-bonding network to the PNIPAM surface of Bound waters, exchange after half the residence time.

In Fig. 1*D*, a decrease in the number of Wrap water molecules correlates with the polymer transition from an extended to a compact state, closely mirroring the change in R_g. By contrast, the Bound water count is relatively invariant (Fig. 1*E*), showing only a weak correlation with R_g. This behavior is consistent with findings from THz spectroscopy studies on liquid–liquid state separation (LLPS) of proteins, which suggest that the release of entropically unfavorable Wrap water facilitates phase separation, whereas the retention of Bound water reduces the enthalpic cost (4).

Besides Wrap and Bound MD observables, additional hydration-related properties such as the solvent-accessible surface area (SASA) for hydrophobic and hydrophilic residues, EPHOA (exposed hydrophobic area), and EPHIA (Exposed hydrophilic area = SASA-EPHOA), were computed to characterize changes in the polymer's exposure to water (see *Methods* for how all of these are computed from MD trajectories).

Reaction Coordinate by Dimensionality Reduction. To visualize transition state passage between extended and compact states (33, 34) and correlate it with solvation, we computed 5 collective coordinates for PNIPAM: R_g, EPHOA (exposed hydrophobic area), EPHIA (Exposed hydrophilic area), Wrap water, and Bound water (defined in *Methods*). We found that two PCA components, PC1 and PC2, extracted by singular value decomposition, were sufficient to account for 93.5% of the variation in trajectories (*SI Appendix*, Table S5). The *Top Left* panel in Figs. 2 and *SI Appendix*, Fig. S8*A* shows a PNIPAM 30-mer trajectory in PC1-PC2 space undergoing a transition state passage (other panels of the user interface in Fig. 2 are explained in the section on "Visualization and Sonification").

Not surprisingly, using only three coordinates—R_g along with either the EPHOA/EPHIA pair or the Wrap/Bound pair—achieves a separation between compact and extended polymer states that is nearly equivalent to that produced by the full set of 5 coordinates (*SI Appendix*, Table S6). This is due to the strong correlation between the two sets of coordinates EPHOA/EPHIA and Wrap/Bound: Wrap water preferentially solvates the exposed hydrophobic areas, while Bound water solvates the polymer's hydrophilic regions. We note that R_g is not necessary to filter compact from extended states, but adding it as a third coordinate to either the EPHOA/EPHIA pair or to the Wrap/Bound pair of coordinates enhances the separation of compact and extended states in the new coordinate system, where PC1 is essentially a measure of compactness, whereas PC2 measures whether a transition state passage along PC1 is dominated by changes in hydrophobic or hydrophilic interactions (*SI Appendix*, Table S6). Hence, we refer to the new coordinate space as C-P space: where the horizontal axis is a measure of *C* (for *C*ompactness, PC1) and the distance along the vertical axis is *P* (for *P*hobo-philic, PC2). Note that R_g and *C* (PC1) are correlated, whereas R_g and *P* (PC2) are not (*SI Appendix*, Fig. S8). Movie S1 shows one trajectory through C-P space alongside the corresponding 3D molecular animation: Compact states are toward the left (yellow), intermediate states are in the center (cyan), and expanded states are toward the right (magenta); a red dot indicates the current position in the state space.

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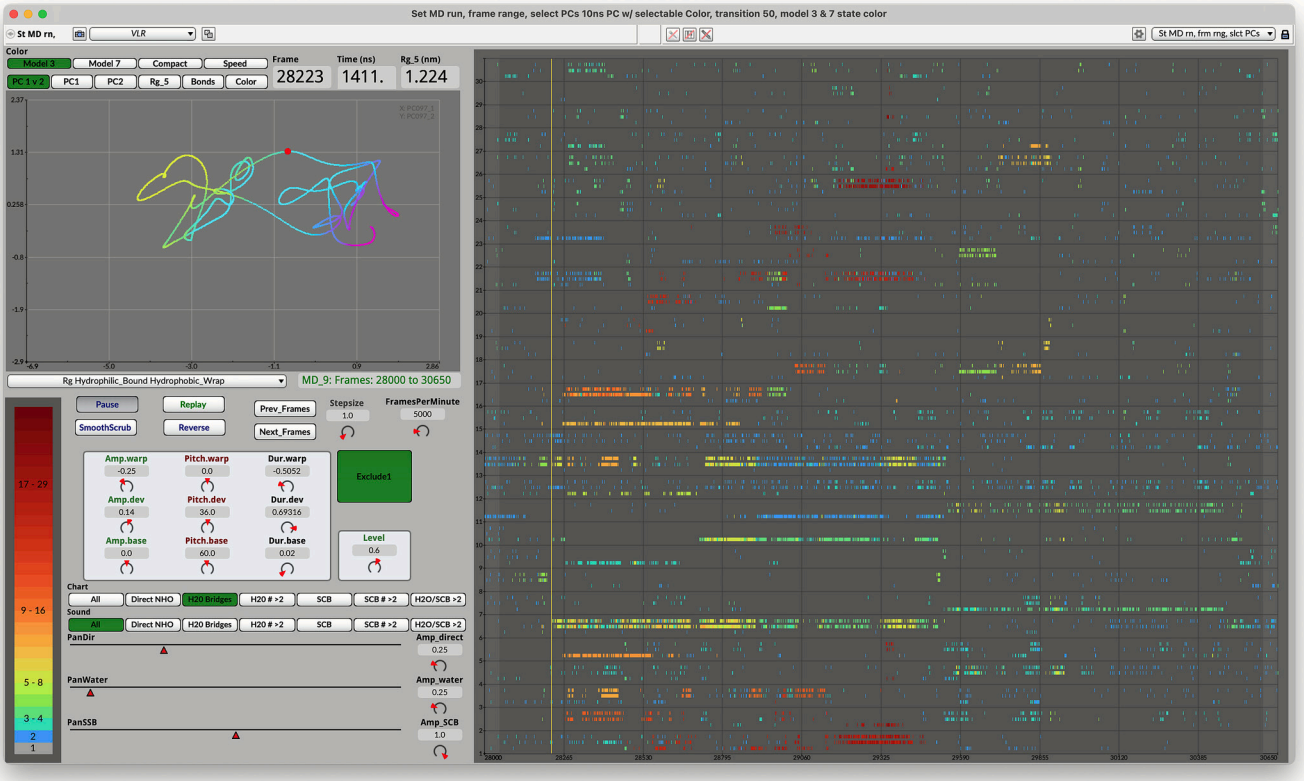


Fig. 2. The sonification/visualization user interface for studying polymer collapse. *Top Left*: This window displays two of the PCA components with the largest singular values (PC1 on the horizontal axis and PC2 on the vertical axis). Toward the left are globule states (yellow), toward the right are coil states (magenta). The red dot indicates the location of the polymer at the frame/time/R_g displayed above the window; in the snapshot PNIPAM is undergoing a transition state passage from extended to compact. Alternatively, the *Top Left* panel can display individual coordinates as a function of time. *Right*: The "piano roll" shows the state of WBs (alternatively, DBs or SCBs) as a function of time (*Horizontal* axis) and NIPAM residue (*Vertical* axis); the yellow bar is the time corresponding to the red dot in the C-P space display, when PNIPAM has just lost its water–water HB network and is beginning to form long-distance WBs. The color bar on the *Bottom Left* codes distance of interactions on the piano roll from local (blue) to highly nonlocal (dark red) by residue separation along the sequence. *Middle Left*: Radio buttons for selecting which bond motifs are displayed on the piano roll and/or heard in the sonification. This makes it possible to follow the reaction coordinate visually and simultaneously follow hydrogen bonding patterns by sound.

Types of H-Bonding Patterns. At the individual bond level of detail, we identified three primary interaction motifs using conventional probability density plots (Fig. 3): direct PNIPAM–PNIPAM N–H···O H-bonds (DB); water-mediated bonds (WB); and side chain–side chain N–H···N interactions (SCB). Probability density $P(x)$ is related to the free energy landscape by PMF(x) = $-RT\ln(P(x))$, where x are the chosen coordinates and PMF is the potential of mean force (PMF) (*SI Appendix*, Fig. S9). The PNIPAM side chain, shown in Fig. 3*A*, is able to form one intramolecular hydrogen bond: N–H to O=C; linear HNH bonds are not observed. All other H-bonds must be to water molecules. Fig. 3*A* shows a histogram of N···O distance vs. N–H···O angle with the expected maximum at about 2.8 ± 0.2 Å and 160 ± 20°; an example is visualized in Fig. 3 *C*, *Left*. However, PNIPAM can form many more water-bridged hydrogen bonds, an example of which is shown in Fig. 3 *C*, *Right*. PNIPAM side chains also interact via another structure, termed here the "side-chain bond" or SCB, characterized by an N–H···N angle of 90 ± 20° and an N···N distance of 3.8 ± 0.2 Å between different side chains (Fig. 3 *B* and *C*). When a series of SCBs forms between adjacent side chains, it can contribute to the long-range organization of the polymer. PNIPAM can also form side chain–side chain interactions via two or three hydrogen bonds bridged by a single water molecule, as seen in Fig. 3 *C*, *Middle*. These WBs greatly outnumber the DBs in all simulations (*SI Appendix*, Fig. S10), confirming the observation by Tamai and coworkers (12), and are similar in importance to the water-mediated interactions in protein–protein recognition (35).

Visualization and Sonification of H-Bond Patterns. Having identified these H-bond motifs, we next investigated their dynamics. While the reduced-dimensional trajectory of PNIPAM is readily visualized (Fig. 2 top left), analyzing H-bonds from a traditional 3D molecular animation (36) is more challenging because many bonds are made and broken in different places and in rapid succession. To address this challenge, we developed an interactive, multimodal display that combines data sonification with two synchronously presented visualizations. This display made it possible to correlate H-bond type, timing, nonlocality, and prevalence with PNIPAM collapse along the 2D projected reaction coordinate discussed above. The multimodal sonification/visualization tool made it possible for us to monitor several time-dependent variables in parallel (Figs. 2, *SI Appendix*, Figs. S8, S11–S13, and Movie S2). The *Top Left* of Fig. 2 shows either the trajectory through C-P state space (as shown in Movie S1) or the time evolution of individual variables such as R_g. On the right, the nonlocality of H-bonds is color-coded in a piano roll, with time as the horizontal axis and residue number as the vertical axis. At the bottom left are controls over the sound parameters. Each H-bonding event is mapped to a sound grain (*SI Appendix*, Fig. S14) whose pitch is set by the residue (and the contact type within each residue) and whose amplitude and duration are mapped to the number of residues spanned by the bond (the nonlocality of the bond); panning across the stereo field helps distinguish DB/WB/SCB, any of which can be selected in different combinations. The tool is highly customizable, with adjustable rate,

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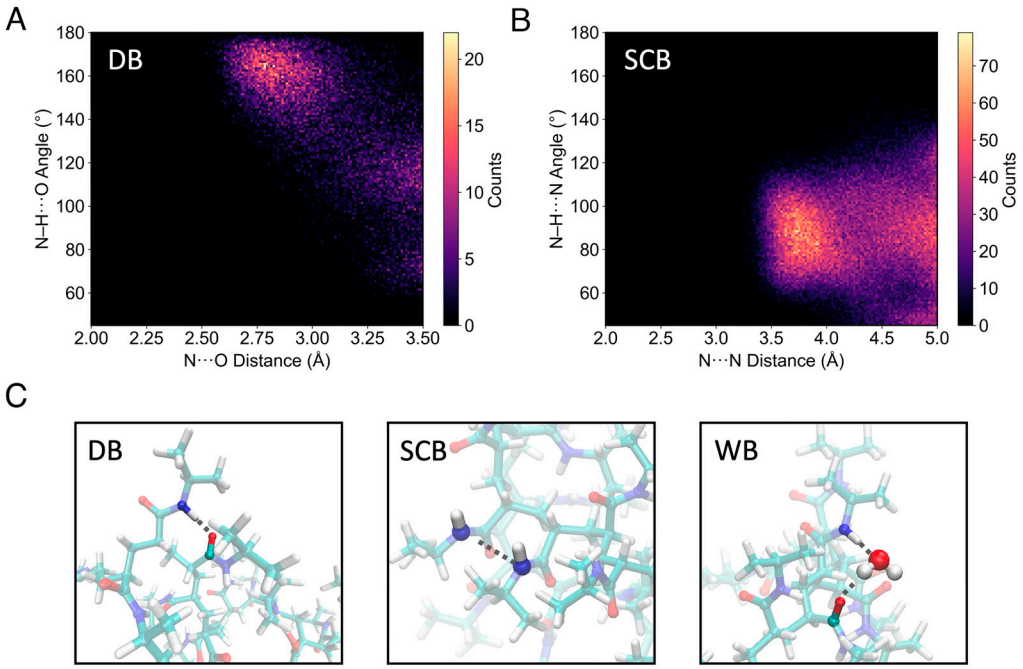


Fig. 3. (*A*) Histogram of direct hydrogen bonds as a function of N···O distance between PNIPAM side chains and N–H···O bond angle. The PNIPAM side chain is shown in the inset. (*B*) A plot of N–H···N angle vs. N···N distance between different subunits reveals a marked population near 90° and 3.7 Å; we term this "side-chain bonding" or SCB. (*C*) Examples of direct inter-side-chain hydrogen bonds (DB), SCB, and water-mediated inter-side-chain hydrogen bonds (WB).

parameter ranges, and perceptual mappings (from log to linear to exponential) (see *SI Appendix*, SI text: Listening to Hydrogen Bonds). Since bonding events by residue (and the contact type—within each residue) are perceptually separated in pitch space and the auditory stream associated with each H-bond motif is spatially separated by stereo position, we could track multiple bond types in parallel. Furthermore, by monitoring one bond type visually while simultaneously monitoring another bond type aurally, we were able to detect examples of cooperativity during polymer collapse that might otherwise have gone undetected.

As can be heard in the example in Movie S3, and also seen in Fig. 2, where the red dot in the top left plot marks PNIPAM just entering the transition passage region toward the globule state, formation and loss of compact states is strongly correlated with formation and loss of long-range WBs (residue spacing >5) DBs are frequently spatially bracketed by WBs and can coexist with WBs on the same residue (*SI Appendix*, Fig. S12), but their formation rarely correlates with the start of collapse or expansion. SCBs can coexist or substitute for WBs (*SI Appendix*, Fig. S12), but they rarely precede collapse of the polymer chain.

Not all transitions are as fast and direct between globule and coil as the ones shown in Figs. 2 or *SI Appendix*, Fig. S8. For example, *SI Appendix*, Fig. S13 shows that PNIPAM can get trapped in a very long-lived (>275 ns) state of intermediate compactness, which coincides with long-lived DBs (red), SCBs (green), and WBs (blue) (Movies S4 and S5). The ensemble of traps within the middle Gaussian in Fig. 1*C* does not correspond to a single well-defined structural state, but rather a multitude of semicompact structures whose hydrogen bond pattern hinders further collapse and must be partly dissolved to reach the more compact globule. A cascade of such traps can occur en route from coil to globule resulting in a "ball hopping down stairs" motion of PNIPAM moving successively lower on the free energy landscape, as in the example sonified and visualized in Movie S6. In these extreme cases, PNIPAM contracts gradually in many small steps from coil to globule, rather than in a single highly cooperative step; the reverse stepwise ascending in energy can also occur (Movie S7). Large single vs. small multistep transitions are also evident in Figs. 1 *B* and *D*. Contrast these multistep transitions with a more direct collapse (Movie S8) and expansion (Movie S9).

The correlation between nonlocal WBs with more than two side-chain contacts and compactness is summarized by the histogram in Fig. 4, where water-mediated H-bonds are color-coded from long range (red) to short range (blue) and structures have been sorted by R_g from globule (*Left*) to coil (*Right*). Similar plots for all WBs, DBs and SCBs are shown in *SI Appendix*, Figs S15–S19, and also visualized in terms of bond counts vs. R_g (*SI Appendix*, Fig. S20). Long-range water-mediated hydrogen bonds (WBs) are most prevalent in compact structures, and their distribution is similar between center and edge residues. The excess of longer-distance SCB interactions (*SI Appendix*, Fig. S17) at mid-range R_g values (~1.22 to 1.5 nm) hints at a possible role for SCBs in trapping the polymer in intermediate states.

Follow-Up Data Analysis. We next performed conventional energy landscape analysis to correlate the DB and WB bond motifs with respect to polymer compactness. *SI Appendix*, Fig. S9 shows 2D and 3D plots of PMF as a function of R_g and the number of bonds of each type. While the simple count of DBs and WBs is weakly correlated with R_g, the distance-weighted WBs show a strong correlation between long-distance WBs and low R_g, confirming that WBs are the primary type of H-bonding that mediates PNIPAM chain interactions during collapse.

Fig. 5 illustrates the local and nonlocal connectivity of PNIPAM in a globule, intermediate, and coil state via a graph network, with numbered NIPAM residues in gray (or green when they form SCBs). Extended states have an extensive network of WBs between adjacent residues, whereas nonlocal WBs predominantly appear in intermediate and compact states, together with a considerable number of SCBs. Nonlocal DBs exist, but are relatively rare even in the most compact states. Compared to intermediate states, compact states

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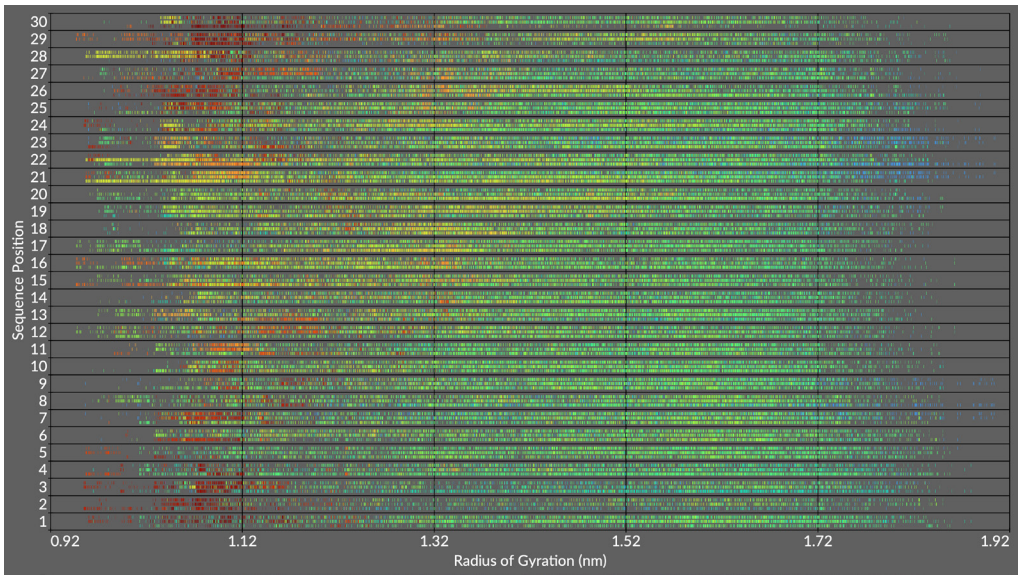


Fig. 4. WBs making greater than two side-chain contacts along the PNIPAM sequence from 1 to 30 (*Vertical* axis), as a function of increasing R_g from 0.92 to 1.92 nm (*Horizontal* axis). H-bonds are color-coded from long range (red) to short range (blue). Long-range WBs are concentrated between R_g 1.1 to 1.2 nm on the *Left*.

exhibit a greater number of nonlocal WBs that also span longer ranges. Most notably, compact states have many centrally located nodes (betweenness 'centrality' is quantified by what fraction of time a node lies on a path connecting other node pairs, see *Methods*).

Fig. 6 illustrates the role of long-range contacts in driving the coil–globule transition. To emphasize distance effects, as discussed previously in the PMF analysis, rather than simply counting WBs and DBs, we weighted each contact by the residue–residue separation $d_{ij} = |i - j|$, where i and j are the connected residues. A higher score reflects longer-range connectivity. Successful transitions are defined by R_g dropping below 1.27 nm (shaded green), whereas failed attempts are shown in red (which may nonetheless yield compact intermediates).

Fig. 6*A* shows that the success of coil–globule transitions strongly correlates with the distance-weighted contact score of both WBs (blue) and DBs (red). In panel A, successful coil–globule transitions are accompanied by a buildup of several nonlocal WBs, whereas DBs occur mainly after the onset of the transition to compactness (Movies S10 and S11). Nonlocal WBs are both more numerous and they span longer ranges, leading to centrally located nodes and increased centrality value, which is size-coded in the graph in Fig. 6*B*. The centrality value increases rapidly as the oligomers pass from coil through the transition state ensemble into the globule state, while failed attempts (occupation of intermediate states) result in small or no gain in centrality (Fig. 6*C*). Importantly, centrality is not significantly enhanced for residues located in the middle of the sequence. Unlike a protein, which forms specific long-range interactions to create a unique folded conformation, in a homopolymer like PNIPAM, side chains near the periphery are as likely as those near the center to serve as important bridges for long-range H-bond contacts, with the exception of the outermost two NIPAM residues (Fig. 6*D*). This behavior is likely due to chain symmetry: While the two ends can easily interact to close a loop, residues in the central region are constrained by the polymer's persistence length, which restricts the formation of local contacts.

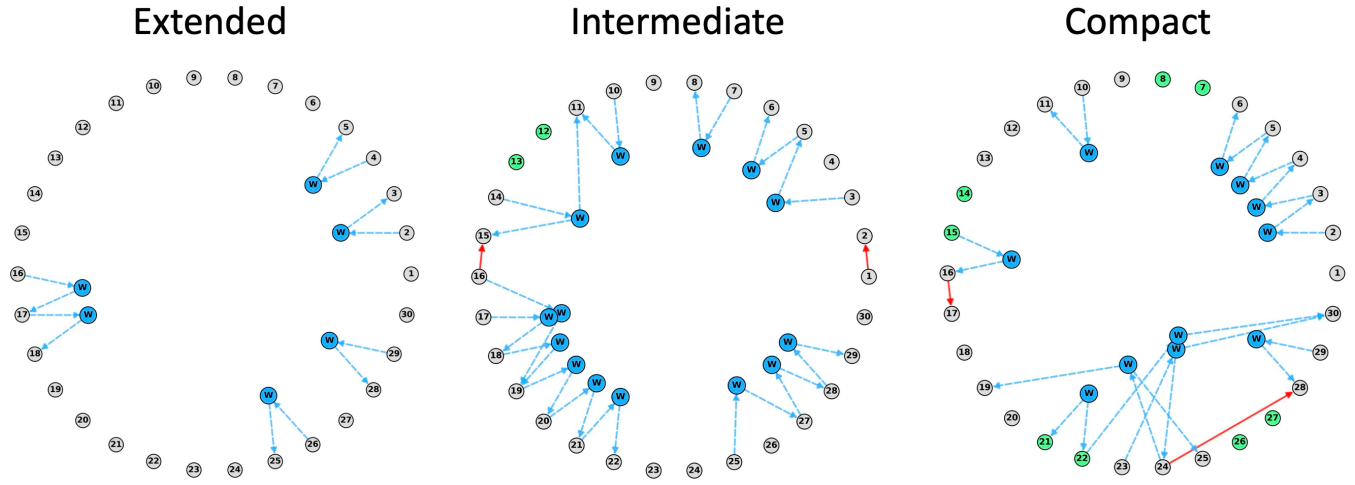


Fig. 5. Representative graphs of a 30-mer PNIPAM chain in extended, intermediate, and compact states (from *Left* to *Right*). PNIPAM residues are shown as nodes numbered from 1 to 30. Edges represent hydrogen bonds, with red solid arrows indicating direct PNIPAM–PNIPAM N–H···O H-bonds (DB), and blue dashed arrows indicating water-mediated H-bonds (WB), where bridging water molecules are shown as blue nodes. Residues involved in side-chain hydrogen bonds (SCB) are highlighted in green. While WB occurs in all three conformational states, long-range DB and WB (connecting nonconsecutive residues) predominantly appear in the compact state.

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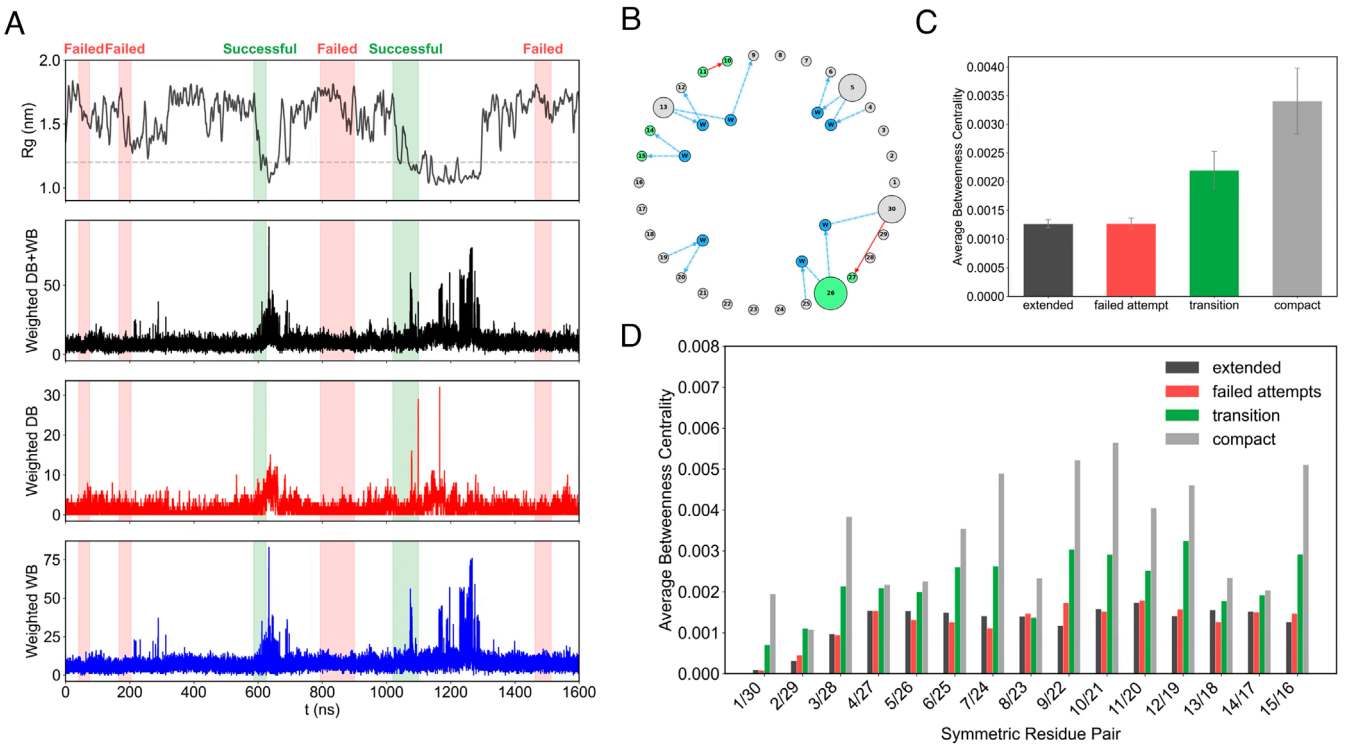


Fig. 6. Long-range contacts play a key role in the PNIPAM globule–coil state transition. (*A*) Time series from a representative MD trajectory showing (*Top* to *Bottom*): radius of gyration (R_g), total distance-weighted direct (DB), and water-mediated (WB) hydrogen bonds among PNIPAM residues, DB only, and WB only. Transitions are defined here by R_g dropping below 1.27 nm; shaded green for successful transitions, red for failed attempts (that may nonetheless lead to compact intermediates). (*B*) Graph of a compact PNIPAM state with node size scaled by centrality, which quantifies what fraction of the simulation a node lies on a path connecting other node pairs—a proxy for structural connectivity. (*C*) Average centrality values across states deduced from all eleven MD trajectories: black (extended), red (failed attempt), green (successful transition), and gray (compact). Centrality is lowest in extended states and failed attempts, and highest in compact states, with transitions showing intermediate values—highlighting the importance of increased connectivity for successful collapse. (*D*) Node-resolved centrality values show a broadly uniform distribution across residues, suggesting no localized hotspot governs the transition.

To further examine the contact range effect, we plot the distribution of WB contact distance d_{ij} in Fig. 7*A*. Near-neighbor contacts ($d_{ij} \leq 5$) are more prevalent than long-range ones ($d_{ij} > 5$). This is expected for PNIPAM 30-mer, because it is a homopolymer with uniformly distributed residues. Nevertheless, long-range contacts that span 5-30 residues do exist as shown by the inset histogram and our sonification/visualization analyses. Near-neighbor and long-range contacts with respect to R_g are shown in Fig. 7 *B* and *C*, respectively. The critical role of long-range WBs on PNIPAM compaction is highlighted by the strong correlation between the number of long-range contacts and R_g, a finding consistent with our sonification/visualization results. In contrast, short-range WBs (Fig. 7*B*) are prevalent even at large coil sizes. This is true for the 60-mer as well (*SI Appendix*, Fig. S5*B*).

Discussion

Although the importance of hydrophobicity in polymer collapse is well understood, the role of water in aqueous polymer compaction is more active and complex than has often been assumed, substantiating the view that "water pulls the strings" (37). Through a combination of MD, auditory analytics, and graph theory, we were able to reveal previously hidden dynamics. Our observations indicate that direct, intrachain hydrogen bonds between amide groups (N–H···O=C) are not the primary structural organizer. Instead, the collapsed globule is organized by a dynamic network of a few persistent, long-distance water bridges, where single water molecules form hydrogen bonds with multiple parts of the polymer chain. We found that the state space of PNIPAM follows

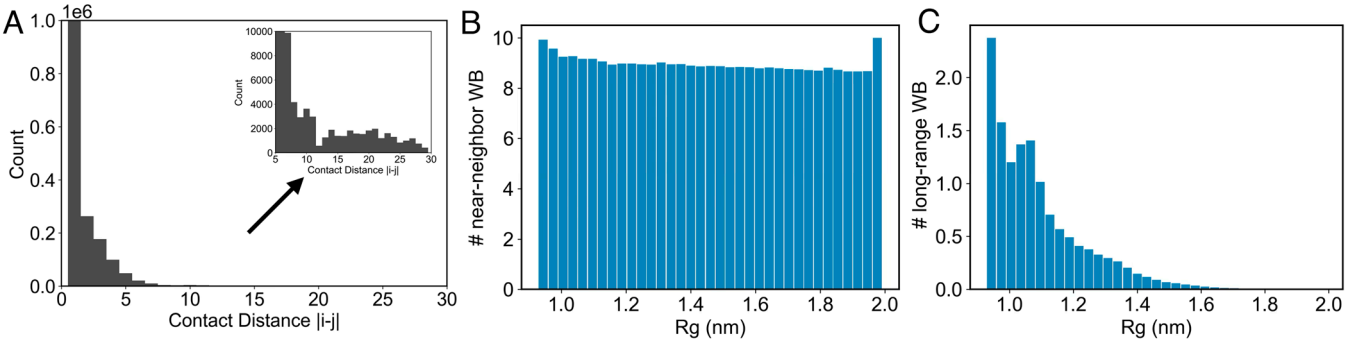


Fig. 7. WB counts by contact distance and R_g. (*A*) Histogram of WB contact distance distribution, with a zoomed-in inset on the long-range contact ($d_{ij} > 5$). (*B*) Average number of near-neighbor WB ($d_{ij} \leq 5$) as a function of R_g. (*C*) Average number of long-range WB ($d_{ij} > 5$) as a function of R_g.

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neither a "direct coil-to-globule" picture, nor a continuous collapse transition, but includes evidence of some direct coil–globule transitions, highly populated intermediate states, and long-lived bonds (mainly WBs and SCBs) that lead either to direct coil–globule transitions or to trapping in intermediate compactness states.

The key result is that water-mediated hydrogen bonds assist hydrophobicity as pathfinders for the globule state, as observed also in proteins (38). The importance of water-mediated structural interactions has also been highlighted for protein–protein recognition (35). The role of DBs, on the other hand, differs between proteins and a homopolymer like PNIPAM: Whereas protein structure relies heavily on backbone H-bonds for secondary organization and salt bridges for long-range tertiary stability, DBs in PNIPAM contribute very little to structural organization. Our observations are in agreement with Yagasaki and Matubayasi (22), who found that the influence of direct polymer–polymer hydrogen bonding on compactness is limited. Thus, to optimize polymer collapse, it is not sufficient to focus solely on tailoring the direct hydrogen bonds: Control over water-mediated contacts is essential.

Methods

Simulation Details. The simulation system consists of a single PNIPAM 30-mer solvated in 32578 water molecules within a cubic box of approximately 10 × 10 × 10 nm^3. The initial PNIPAM configuration was taken from the Supporting Information of Ref. 39. To examine chain length effects, we ran an additional simulation: a 60-mer solvated in 63377 water molecules in a 12.5 × 12.5 × 12.5 nm^3 box (approximately twice the volume of the 30-mer box). Both simulations employed a modified OPLS-AA force field (22, 24) in conjunction with the SPC/E water model (25). This combination was chosen because it more accurately reproduces the demixing temperature and the coil–globule transition temperature of PNIPAM than the original OPLS-AA model (25). For a review of water force fields for PNIPAM, including the SPC/E and OPLS-AA combination, see Tavagnacco et al. (28).

All simulations were performed using the GROMACS package (40). The leapfrog algorithm with a 2 fs timestep was used for integration. Long-range electrostatics were treated with the particle mesh Ewald method, and a standard long-range correction was applied for Lennard–Jones interactions. Covalent bonds involving hydrogen in PNIPAM were constrained using LINCS (41), while SETTLE (42) was used for water. Lorentz−Berthelot mixing rules were applied to determine the interaction between different atom types. Simulations were run in the NPT ensemble at 1 atm and 310 K using the Nosé–Hoover thermostat (43, 44) and Parrinello–Rahman barostat (45). Following energy minimization and 20 ns equilibration, independent production runs were conducted for the PNIPAM 30-mer (11 runs, totaling ~16 μs) and the PNIPAM 60-mer (10 runs, totaling ~8 μs).

Three specific interactions–intra-PNIPAM hydrogen bonds (N–H···O), PNIPAM-water hydrogen bonds, and side-chain nitrogen–nitrogen interactions (N–H···N)–were identified and defined via histogramming (Fig. 3), as detailed in the Results. For example, based on the PNIPAM-water histograms, cut-offs of >140° (angle) and <3 Å (distance) were chosen to capture ≥95% of the distribution. These criteria align with cut-offs proposed in the literature for "moderate, mostly electrostatic" hydrogen bonds (46).

Analysis details. Following a simple Gaussian fit to the R_g distribution (*SI Appendix*, Fig. S3 and Table S1), we developed the fast Kinetic Transition Estimator (KiTE), an $O(kN \log N)$ algorithm (where N = number of data frames, k = number of states), and applied it to the raw R_g data to estimate the start times and durations of coil–globule transitions in the simulation. The transition durations identified by KiTE are related by a scale factor to the transit times found using the STaSI method with logistic fits (29, 30). Further details on the KiTE and STaSI algorithm and fits are available in the Supporting Information (*SI Appendix*, Figs. S1, S2, S4, S5, and S21 and SI text). Representative transitions identified by these methods are shown in *SI Appendix*, Tables S2 (STaSI) and S3 (KiTE).

Collective variables were defined as follows:

$$R_g = \frac{1}{M} \sum_i m_i r_i^2, \quad [1]$$

where R_g is the radius of gyration, M total PNIPAM mass, m_i individual atom masses, and r_i individual atom distances from the PNIPAM center of mass. The solvent-accessible surface area (SASA) was computed using the FreeSASA package (47), as the van der Waals surface of the solute expanded by a spherical solvent probe (radius 0.14 nm) centered on each atom, with van der Waals radii taken from Bondi (48, 49). The resulting SASA values were then partitioned into contributions from hydrophilic (N and O) and hydrophobic (methyl C) residues to obtain the exposed polar (EPHIA) and nonpolar (EPHOA) surface areas, respectively. The R_g time series was also analyzed using a conventional autocorrelation analysis (*SI Appendix*, Fig. S6).

Polymer hydration was characterized by identifying water molecules within the first solvation shell of PNIPAM, using the definitions of cavity "Wrap" and "Bound" water (4, 50–54) as showcased in Ref. 54 for the glycerol/water system. Wrap water surrounds nonpolar methyl groups (within 5.5 Å of a methyl carbon) without direct H-bonding to the polymer, forming a two-dimensional water–water HB network wrapped around the polymer. This network shows a characteristic THz signature at 150 to 160 cm^{-1} due to collective HB stretching, red-shifted relative to bulk water. Bound water forms direct H-bonds with PNIPAM, defined as water whose oxygen lies within 3.5 Å of a PNIPAM nitrogen or oxygen atom, with THz signature in the librational frequency range (400 to 600 cm^{-1}) (4). The numbers of both Wrap and Bound water were normalized by the number of PNIPAM repeat units. The H-bond connectivity and different water populations (Wrap, Bound, and bulk water) are shown and discussed in *SI Appendix*, Fig. S7 and Table S4.

Hydrogen bond dynamics were studied using auditory analytics (55), an approach that uses data sonification and visualization to reveal features that might otherwise escape detection; all features are subsequently confirmed using conventional statistical data analysis (21). We designed two primary tools for studying H-bond dynamics. First, a 2D state space for PNIPAM was created using principal component analysis, reducing several collective coordinates smoothed by a weighted time-average window to a minimal set of two reaction coordinates. The trajectory of PNIPAM through state space can be followed visually while sound cues indicate the times when various types of H-bonds form. Second, for studying individual trajectories, we designed an interactive piano roll: a sonification and visualization of bond formation with respect to time and repeat unit along the chain, which we displayed in conjunction with the state space trajectory. Several parameters of the multimodal display, including bond type, sequence separation or "span" (distance between donor and acceptor by residue number), sound parameter ranges, rate, and reaction coordinate, are adjustable in real time, allowing the listener to focus on various aspects of the data. These multimodal displays facilitated the identification of spatially distributed patterns of H-bonding–complex time-evolving patterns that were considerably more difficult to discern when visualizing traditional 3D MD simulation animations.

These tools were developed using the Kyma language (56) [a sound design framework written in the Smalltalk programming language (57)] and are an extension of tools developed for studying H-bond dynamics during the protein-folding reaction (21). See Fig. 2 and *SI Appendix*, Figs. S8 and S11–S14, and SI text for additional details on data shaping, sound synthesis methods, and parameter mapping.

To characterize the overall correlation of bond types with compactness of the polymer, histograms were created to show each bond type, color-coded by sequence distance, as a function of sequence position (residue number) and R_g. To further visualize bonding patterns between side chains, we created a graph topology where the residues are nodes, and bonds as type-coded edges between nodes. Finally, betweenness centrality was used to highlight which residues are "central" to the network of bonds, i.e. are links with maximal number of connections to other residues by measuring the fraction of shortest paths passing through a given residue. The betweenness centrality of residue $\subset$ is defined as:

$$C_B(v) = \sum_{s \neq v \neq t} \sigma_{st}(v)/\sigma_{st}, \quad [2]$$

where σ_{st} is the total number of shortest paths between residues s and t, and $\sigma_{st}(\subset)$ is the number of those paths that pass through $\subset$. Residues with high betweenness centrality serve as central "hubs" in the bonding network.

Data, Materials, and Software Availability. All Study data are included in the article and/or supporting information.

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